IBM Data Science Capstone Project

SpaceX Falcon 9 Launch Success Prediction using Machine Learning

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Executive Summary

This project predicts whether the SpaceX Falcon 9 first-stage landing will be successful. The dataset was collected from the SpaceX API and Wikipedia. Key features include payload mass, orbit type, and launch site. After Exploratory Data Analysis (EDA) and classification modeling, Logistic Regression achieved the best accuracy (≈83%).

Introduction

SpaceX aims to reduce launch costs through reusable rockets. This project investigates which factors influence successful landings using open data sources. The analysis follows a structured pipeline from data collection to machine learning prediction.

Data Collection and Wrangling

Data was collected via SpaceX REST API and Wikipedia HTML tables using BeautifulSoup. Data cleaning involved removing Falcon 1 launches, filling missing payload values with mean, and creating a binary label 'landing_class'.

Exploratory Data Analysis (EDA) Methodology

EDA used pandas, seaborn, matplotlib, and SQL queries to analyze launch outcomes. Visualizations were produced for payload mass vs. success, site performance, and orbit success rates. Folium was used for interactive mapping and Plotly Dash for real-time dashboards.

EDA Results using SQL

SQL queries provided insights such as: • Total launches by site (KSC LC-39A highest success rate ≈77%)
• Average payload per orbit
• Success proportion by launch site and year

EDA Visualization Highlights

Charts revealed:

• Positive correlation between payload mass (4000–6000 kg) and landing success
• LEO orbits showed the highest reuse success rate
• Increasing yearly success trends since 2013

Interactive Map using Folium

Mapped all launch sites, marked closest coastline distances, and visualized launch routes. Insight: Sites near coastlines had higher success rates for drone-ship landings.

Interactive Dashboard using Plotly Dash

Dashboard included site dropdowns and payload sliders to dynamically visualize success rates and scatter plots. Enabled users to explore launch performance by site and payload range in real time.

Predictive Analysis Methodology

Classification algorithms tested: Logistic Regression, SVM, Decision Tree, and KNN. GridSearchCV (cv=10) was used for hyperparameter tuning. Model accuracy was evaluated on test data.

Model Accuracy Results

Model	Accuracy (%)
Logistic Regression	**83**
SVM	78
Decision Tree	75
KNN	77

Logistic Regression performed best overall with balanced precision and recall.

Insights and Interpretation

- Payload mass and orbit are significant predictors of success
- Logistic model explained launch behavior accurately
- Workflow generalizable to future mission data
- Supports SpaceX in improving reusability strategy

Conclusion

This project successfully developed a machine learning workflow for predicting Falcon 9 first-stage landing success. Interactive dashboards, mapping, and predictive analytics provided valuable insights. Future work: integrate weather and engine telemetry for enhanced prediction.

GitHub Repository

All notebooks, datasets, and dashboards available at:
https://github.com/yourusername/falcon9-prediction
Includes: EDA.ipynb, SQL_Analysis.ipynb, Dashboard.ipynb, ML_Prediction.ipynb.

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